ROC Curve & AUC Evaluation Report

SVM – Breast Cancer Classification

Objective:

The objective of this evaluation is to measure the classification performance of the trained **Support Vector Machine (SVM)** model using:

- Receiver Operating Characteristic (ROC) Curve
- Area Under the Curve (AUC) Score

These metrics help determine how well the model distinguishes between:

- **Malignant tumors**
- **Benign tumors**

Methodology:

1. Used the **best tuned SVM model** obtained from GridSearchCV.
2. Generated prediction probabilities using:
   - predict_proba()
3. Calculated:
   - True Positive Rate (TPR)
   - False Positive Rate (FPR)
4. Plotted ROC curve using Matplotlib.
5. Computed AUC score using:
   - sklearn.metrics.auc

Results:

ROC Curve

The ROC curve plots:

- X-axis → False Positive Rate
- Y-axis → True Positive Rate

The curve remained close to the **top-left corner**, indicating strong model performance.

AUC Score

- AUC Score ≈ 0.99

Interpretation:

- AUC close to **1.0** indicates excellent discrimination ability.
- Model effectively separates malignant and benign tumors.
- Very low false positive and false negative rates.
- Performance is significantly better than a random classifier (AUC = 0.5).

Key Observations:

- Feature scaling improved SVM accuracy.
- RBF kernel handled non-linear relationships effectively.
- Hyperparameter tuning increased generalization performance.
- ROC curve confirmed robustness of the trained model.

Conclusion

The tuned SVM classifier achieved excellent performance in breast cancer classification.

✔ High ROC-AUC score

✔ Strong predictive capability

✔ Reliable model for binary medical classification tasks

ROC-AUC evaluation validates that the model performs accurately and consistently.

6 - Interview Questions – Ready Answers

1. What is the use of Linear Regression?

To model and predict a continuous dependent variable based on independent variables.

2. What is RMSE and why is it important?

RMSE measures average prediction error and penalizes large errors more than MAE.

3. Why do we split train and test data?

To evaluate model performance on unseen data and avoid overfitting.

4. What does a coefficient represent?

Change in target value for a one-unit change in the feature.

5. What is underfitting?

When the model is too simple and fails to capture data patterns.

7-Interview Questions – Crisp Answers

Why Logistic Regression is used for classification?

It predicts probability and applies a threshold for class labels.

What is precision and recall?

Precision = correctness of positive predictions

Recall = ability to find actual positives

What is ROC curve?

Plot of TPR vs FPR at different thresholds.

What does AUC mean?

Overall ability of model to distinguish classes.

What is confusion matrix?

Table showing TP, FP, FN, TN counts.

Interview Questions – Ready Answers

Why is accuracy misleading in fraud detection?
Because fraud cases are rare and accuracy ignores minority class performance.

What is Random Forest?
An ensemble of decision trees using bagging and feature randomness.

What is ensemble learning?
Combining multiple models to improve performance.

What is n_estimators**?**
Number of trees in the forest.

What is SMOTE?
A technique to oversample minority class by generating synthetic samples.

Evaluation Report – Decision Tree Model

Dataset Used

The model was trained and evaluated using the **Kaggle Bank Marketing Dataset**, which contains customer-related attributes to predict whether a client will subscribe to a term deposit.

Model Performance

The Decision Tree Classifier was evaluated using **training accuracy** and **testing accuracy** to assess both learning capability and generalization performance.

Training Accuracy: 0.7900

Testing Accuracy: 0.7721

Analysis of Results

The training and testing accuracies are **very close**, with a difference of approximately **2%**, indicating that the model generalizes well to unseen data.

There is **no significant overfitting**, as the model does not show a large performance drop on the test dataset.

The slightly lower accuracy is expected because:

- The dataset is **imbalanced**, with more "no" subscription cases.
- The decision tree depth was intentionally limited to avoid overfitting and maintain interpretability.

Overfitting Check

Since the testing accuracy closely follows the training accuracy, the model demonstrates **stable and balanced performance**. This confirms that controlling the max_depth parameter effectively reduced overfitting.

Conclusion

The Decision Tree model achieved **reasonable and reliable accuracy** on both training and testing data. The results indicate that the model is suitable for understanding customer subscription behavior and provides interpretable decision rules, fulfilling the objectives of the task.